

Overview over the notes on polyhedra

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1 Notation and concepts

In all notes $\mathbf{A}$ denotes an $m \times n$ matrix, and $\mathbf{b}$ denotes an $m \times 1$ vector. The set $\mathbf{M} := \{1, \dots, m\}$ is used to index the rows of A , and the rows of A are denoted $\mathbf{a}_i$ for $i \in M$.

A **polyhedron** is a set of the form $\{x \in \mathbb{R}^n : Ax \leq b\}$, *i.e.*, the solution set to a finite number of inequalities.

Given points $x^1, \dots, x^k \in \mathbb{R}^n$ and scalars $\mu \in \mathbb{R}^k$, the point:

$$\bar{x} := \mu_1 \cdot x^1 + \dots + \mu_k \cdot x^k$$

is called a **linear combination** of $x^1, \dots, x^k$. If $\mu \geq 0_k$, then $\bar{x}$ is called a **conic combination** of $x^1, \dots, x^k$. If $\bar{x}$ is a conic combination of $x^1, \dots, x^k$ and $\mu_1 + \dots + \mu_k = 1$, then $\bar{x}$ is called a **convex combination** of $x^1, \dots, x^k$.

A set is called **linear** if it contains all linear combinations of its elements, a **convex cone** if it contains all conic combinations of its elements, and **convex** if it contains all convex combinations of its elements. A set is called a **polyhedral cone** if it is both a convex cone and a polyhedron.

The following notation is used. Given $S \subseteq \mathbb{R}^n$:

$\text{span}(\mathbf{S}) \sim$ set of all linear combinations of points in S .

$\text{cone}(\mathbf{S}) \sim$ set of all conic combinations of points in S .

$\text{conv}(\mathbf{S}) \sim$ set of all convex combinations of points in S .

The set $\text{cone}(\mathbf{S})$ is a convex cone (proved in note 1.(A)), and the set $\text{conv}(\mathbf{S})$ is a convex set (proved in note 2.(A)).

A set of det form $Q = \text{conv}(V) + \text{cone}(E)$ with $V, E \subset \mathbb{R}^n$ finite is called **finitely generated**. If $V = \emptyset$, then Q is called a **finitely generated cone**.

Points in a convex cone can be viewed as **directions**. For a direction the important property is where it points, and not its length. Two directions $d^1, d^2 \in \mathbb{R}^n$ are therefore **equivalent** if one has $d^1 = \alpha \cdot d^2$ for some $\alpha > 0$.

Given a convex set $K \subseteq \mathbb{R}^n$, an **extreme point** of K is a point in K that is **not** a convex combination of other points in K .

Given a convex cone $K \subseteq \mathbb{R}^n$, an **extreme direction** of K is a direction $d \in K \setminus \{0_n\}$ that is **not** a conic combination of other (non-equivalent) directions in K .

For an extreme direction $d \in K$, the set $R(d) := \{\alpha \cdot d : \alpha > 0\}$ is called the **extreme ray** associated with d (the directions equivalent to d).

A convex cone $K \subseteq \mathbb{R}^n$ can contain a non-trivial linear space, *i.e.*, a linear space $L \subseteq K$ that does not only consist of the origin. This is the case precisely when the cone K contains a line. A convex cone is called **pointed** if it does *not* contain a line.

Given a set $S \subseteq \mathbb{R}^n \times \mathbb{R}^k$, the **projection of S onto $\mathbb{R}^n$** is the set:

$$\text{Proj}(S) := \{x \in \mathbb{R}^n : \text{there exists } y \in \mathbb{R}^k \text{ such that } \begin{pmatrix} x \\ y \end{pmatrix} \in S\}$$

(points $x \in \mathbb{R}^n$ that can be extended with some $y \in \mathbb{R}^k$ such that $\begin{pmatrix} x \\ y \end{pmatrix} \in S$).

An inequality $\alpha^T x \leq \beta$ is **valid** for a set $S \subseteq \mathbb{R}^n$ if $\alpha^T x \leq \beta$ holds for all points $x \in S$. If $\alpha^T x \leq \beta$ is valid for S , and also $\alpha^T \bar{x} = \beta$ for some point $\bar{x} \in S$, then $\alpha^T x \leq \beta$ is called **supporting** for S .

All supporting inequalities for a convex cone have zero on the right-hand-side (proved in note 1.(B)). The set of normal vectors of supporting inequalities for a convex cone $K \subseteq \mathbb{R}^n$ is called the **dual cone** of K and is denoted:

$$K^\circ := \{\alpha \in \mathbb{R}^n : \alpha^T x \leq 0 \text{ is valid for } K\}$$

For a system $Ax \leq b$ and a point $\bar{x} \in \mathbb{R}^n$, let $\mathbf{I}(\bar{x}) := \{i \in M : a_i^T \bar{x} = b_i\}$ index those inequalities of $Ax \leq b$ that are satisfied with equality by $\bar{x}$. A point $\bar{x} \in \{x \in \mathbb{R}^n : Ax \leq b\}$ is called a **basic feasible solution** for $Ax \leq b$ if $\bar{x}$ is the **unique** solution to the equalities:

$$a_i^T x = b_i \quad \text{for } i \in I(\bar{x}).$$

For a convex set $K \subseteq \mathbb{R}^n$ there might be directions $d \in \mathbb{R}^n$ along which one can start at an **any** point in K and move **infinitely far** along d without leaving K , *i.e.*, directions d satisfying $\{x + \alpha d : \alpha \geq 0\} \subseteq K$ for **all** $x \in K$. Directions d with this property are called **recession directions** for K .

The convex cone consisting of all recession directions for K is denoted $\text{rec}(\mathbf{K})$ and called the **recession cone** associated with K :

$$\text{rec}(\mathbf{K}) := \{d \in \mathbb{R}^n : \{x + \alpha d : \alpha \geq 0\} \subseteq K \text{ for all } x \in K\}.$$

In note 2.(C) it is shown that the recession cone of a polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is given by:

$$\text{rec}(\mathbf{P}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \leq \mathbf{0}_m\},$$

and that the recession cone of a finitely generated set $Q = \text{conv}(V) + \text{cone}(E)$ is given by:

$$\text{rec}(\mathbf{Q}) = \text{cone}(\mathbf{E}).$$

Almost all results on general polyhedra are obtained by **generalizing results on polyhedral cones** to general polyhedra. This is done via the so-called **cone extension** of a convex set. The cone extension of a convex set is a convex cone that **inherits structure** of the original convex set.

The cone extension $K^{\text{ext}} \subseteq \mathbb{R}^{n+1}$ of a convex set $K \subseteq \mathbb{R}^n$ is obtained by **adding an extra non-negative variable** $x_{n+1} \geq 0$. Specifically the cone extension K^{ext} of K is the smallest convex cone with the properties:

$$\begin{pmatrix} x \\ 1 \end{pmatrix} \in K^{\text{ext}} \iff x \in K \quad \text{and} \quad \begin{pmatrix} x \\ 0 \end{pmatrix} \in K^{\text{ext}} \iff x \in \text{rec}(K).$$

It follows that the original convex set K is available at "level 1" of K^{ext} , and the recession cone of K is available at "level 0" of K^{ext} .

The **cone extension of a polyhedron** $P := \{x \in \mathbb{R}^n : Ax \leq b\}$ is denoted $C_p(A, b)$ and given by:

$$C_p(A, b) := \left\{ \begin{pmatrix} x \\ x_{n+1} \end{pmatrix} \in \mathbb{R}^{n+1} : Ax \leq x_{n+1} \cdot b \quad \text{and} \quad x_{n+1} \geq 0 \right\},$$

and the **cone extension of a finitely generated set** $Q := \text{conv}(V) + \text{cone}(E)$ is denoted $C_f(V, E)$ and given by:

$$C_f(V, E) := \text{cone} \left(\left\{ \begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V \right\} \right) + \text{cone} \left(\left\{ \begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E \right\} \right).$$

2 Description of the notes

The notes are divided in two. Results on polyhedral cones are presented in the first part (notes 1.(A)-1.(H)), and results on general polyhedra are presented in the second part (notes 2.(A)-2.(I)). The main results are:

- 1.(A) The **conic hull** $\text{cone}(S)$ of any set $S \subseteq \mathbb{R}^n$ is a **convex cone**, *i.e.*, $\text{cone}(S)$ contains all conic combinations of its elements.
- 1.(B) All supporting inequalities for a convex cone K have **zero on the right-hand-side**, *i.e.*, an inequality $\alpha^T x \leq \beta$ is supporting for $K \iff \alpha^T x \leq 0$ is valid for K and $\beta = 0$. This implies **every polyhedral cone is of the form** $\{x \in \mathbb{R}^n : Dx \leq 0_p\}$ for some matrix $D \in \mathbb{R}^{p \times n}$.
- 1.(C) **Projections of polyhedral cones are also polyhedral cones**, *i.e.*, the set $\text{Proj}(C)$ is **also** a polyhedral cone when C is a polyhedral cone.
- 1.(D) **Finitely generated cones are also polyhedral cones**, *i.e.*, a finitely generated cone is the solution set to a finite number of inequalities.
- 1.(E) The dual cone of a polyhedral cone $C = \{x \in \mathbb{R}^n : Ax \leq 0_m\}$ is given by $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$, and the dual cone of the dual cone is the original cone, *i.e.*, one has $(C^\circ)^\circ = C$.
- 1.(F) **Finitely generated cones and polyhedral cones are the same objects**. In 1.(D) it was shown finitely generated cones are polyhedral cones. Here it is shown that polyhedral cones are finitely generated.
- 1.(G) A polyhedral cone C is pointed **precisely** when the origin is an extreme point of C .
- 1.(H) A pointed polyhedral cone $C = \{x \in \mathbb{R}^n : Ax \leq 0_m\}$ can be **finitely generated by its extreme directions**, *i.e.*, one has $C = \text{cone}(E^*)$, where $E^* \subset \mathbb{R}^n$ is a finite set of extreme directions for C .
- 2.(A) Convex combinations have the following two properties:
 - (i) **The convex hull** $\text{conv}(S)$ of any set $S \subseteq \mathbb{R}^n$ is a **convex set**, *i.e.*, $\text{conv}(S)$ contains all convex combinations of its elements.
 - (ii) A point $x \in K$ is an extreme point of a convex set $K \iff x$ is **not a convex combination of TWO points in** $K \setminus \{x\}$.

- 2.(B)** Polyhedra and finitely generated sets are the same objects, *i.e.*, every polyhedron $P \subseteq \mathbb{R}^n$ can be written $P = \text{conv}(V) + \text{cone}(E)$ with $V, E \subset \mathbb{R}^n$ finite, and every finitely generated set is the solution set to a finite number of linear inequalities.
- 2.(C)** A polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ has recession cone $\text{rec}(\mathbf{P}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \leq \mathbf{0}_m\}$, and a finitely generated set $Q = \text{conv}(V) + \text{cone}(E)$ has recession cone $\text{rec}(\mathbf{Q}) = \text{cone}(\mathbf{E})$.
- 2.(D)** Relation between a convex set K and its conic extension K^{ext} :
- (i) The recession cone $\text{rec}(K)$ of K is pointed $\iff K^{\text{ext}}$ is pointed,
 - (ii) $x \in K$ is an extreme point of $K \iff \begin{pmatrix} x \\ 1 \end{pmatrix}$ is an extreme direction of K^{ext} ,
 - (iii) $\alpha^T x \leq \beta$ is valid for $K \iff \alpha^T x \leq x_{n+1} \cdot \beta$ is valid for K^{ext} .
- 2.(E)** A polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$, where $\text{rec}(P)$ is pointed, is of the form $\mathbf{P} = \text{conv}(\mathbf{V}^*) + \text{cone}(\mathbf{E}^*)$, where $V^* \subset \mathbb{R}^n$ is the finite set of extreme points of P , and $E^* \subset \mathbb{R}^n$ is a finite set of extreme directions of $\text{rec}(P)$. Furthermore one has $\text{cone}(\mathbf{E}^*) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \leq \mathbf{0}_m\}$.
- 2.(F)** In this note the valid inequalities for a polyhedron are characterized. This generalizes dual cones that characterize the valid inequalities for a polyhedral cone.
- 2.(G)** How can one prove that a polyhedron is empty? This question is answered by the so-called called **Farkas lemma** presented and proved in this note.
- 2.(H)** Here it is shown that **basic feasible solutions are the same objects as extreme points**.
- 2.(I)** The following question is asked: **Given vector $\alpha \in \mathbb{R}^n$ and polyhedron $\mathbf{P}$, for which β is the inequality $\alpha^T x \leq \beta$ supporting for $\mathbf{P}$? Two different linear optimization problems are derived both giving this value. The two problems are called **dual problems**.**